MAHESH BUNAGE

DATA ENGINEER

MSC ARTIFICIAL INTELLIGENCE | STAMP 1G | GALWAY

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PROFILE

- Data Engineer with 2.5+ years at Experian (via Caspex Tech), a global credit and financial data company, in a team running ~109 production pipelines that process terabytes of data daily.
- SQL and PL/SQL (stored procedures, query optimisation), with performance tuning on terabyte-scale Hive tables, ETL development and data warehousing on Hive and Snowflake.
- Experienced in testing (pytest, unittest), Jenkins CI/CD, code reviews, production support and Agile delivery, including a regulatory (CCPA) compliance pipeline with field-level encryption.
- MSc in Artificial Intelligence, with Python, AWS and data API experience.
- **Work Authorisation:** Stamp 1G (Third Level Graduate Programme) from December 2026 – full-time work authorisation in Ireland with no employment permit or sponsorship required.

TECHNICAL SKILLS

- **SQL & Databases:** SQL, PL/SQL (stored procedures), Spark SQL, HiveQL, query performance tuning; Snowflake, PostgreSQL, MySQL, HBase, DynamoDB
- **ETL & Data Warehousing:** ETL pipeline development, data warehousing (Hive, Snowflake), PySpark, Apache Airflow, Sqoop, Hadoop (HDFS, YARN)
- **Data Modelling:** Relational database design (primary/foreign keys, constraints, indexing), Hive DDL, partitioning and bucketing
- **Programming & Scripting:** Python (Pandas, NumPy), SQL, Linux/Unix, cron
- **Cloud & APIs:** AWS (S3, EMR, Glue, Lambda, Athena, Step Functions, SNS, DynamoDB, Parameter Store), Terraform, FastAPI data APIs
- **Engineering Practices:** Unit testing (pytest, unittest), Jenkins CI/CD, code reviews, Git, GitHub, Bitbucket, Agile/Scrum, JIRA, Confluence
- **File Formats:** Parquet, ORC, CSV, JSON, .gz

WORK EXPERIENCE

Dec 2022 – Jul 2025 | Caspex Tech LLP **Associate Software Engineer**

Consultant for Experian

- Part of a data engineering team running ~109 production Airflow pipelines that process terabytes of data daily for Experian, a global credit and financial data company.

CCPA Compliance Data Pipeline | PySpark, Airflow, AWS EMR, S3, Lambda, SNS, DynamoDB, Terraform

- Developed and maintained PySpark jobs on AWS EMR to process marketing data for CCPA (California Consumer Privacy Act) compliance: converting .gz files to Parquet and running search, delete and file-reuse operations on consumer records.
- Orchestrated the pipeline with Airflow DAGs triggered by AWS Lambda, tracked load status in DynamoDB, and sent success/failure alerts through SNS, provisioning the Lambda triggers and SNS topics with Terraform.
- Applied field-level encryption to sensitive fields, and built exception handling and logging to keep the pipeline stable and auditable.
- Wrote unit tests with pytest and unittest, took part in code reviews in Bitbucket, and used Jenkins CI/CD to deploy code to S3 automatically.

Data Warehousing, Tuning & Reporting | Hive, Spark SQL, Snowflake, SQL, S3 Select

- Tuned Hive queries on terabyte-scale tables with partitioning, bucketing and Parquet/ORC; PySpark jobs pull the required columns in about 5 minutes.
- Loaded and transformed large structured and semi-structured datasets (CSV, Parquet, .gz, JSON) from Hive and multiple teams' S3 buckets, and loaded processed data into Snowflake for analytics and reporting.

- Built a weekly S3 storage-cost report pipeline using complex SQL, S3 Select and the S3 API, emailed to stakeholders as an HTML report.
- Provided production support for data pipelines, investigating and resolving failures and data issues, and onboarded 3–4 new enterprise clients.
- Delivered in Agile/Scrum with cross-functional teams, using JIRA and Confluence.

Sayu Softech Private Limited, Pune **Data Engineer Intern**

Domain: Telecom

- Built data-driven reports, wrote stored procedures and optimised queries using SQL and PL/SQL.
- Enterprise data warehouse project: loaded enterprise data from AWS RDS (PostgreSQL) into HBase using Spark and Phoenix, created Hive tables with DDL, de-duplicated the data and delivered results to S3 for reporting.
- Imported data from MySQL into HDFS and Hive using Sqoop, and worked across Hadoop, S3 and AWS Glue in a data lake.

P R O J E C T S

Hotel Reservations Database, ETL & Data API | PostgreSQL, SQL, Python, Playwright, FastAPI

- Designed a relational PostgreSQL schema: a reservations fact table with foreign keys to 3 lookup tables, a unique constraint to prevent duplicates, and 5 indexes for common query paths.
- Built the ETL to scrape 250 reservations with Playwright, clean and type-cast them, and upsert into PostgreSQL, with 6 automated data verification checks.
- Exposed the data through a FastAPI data API with 10 SQL-backed queries (revenue, channel mix, cancellations, pickup) and a SELECT-only safe-SQL endpoint; deployed on Render.

Dublin Bikes Data Analysis | Python, Pandas, Scikit-learn

- Analysed 158,574 Smart Dublin records across 117 stations, finding demand peaks at 7–9 AM and 5–7 PM, and predicted availability with 86.1% accuracy (Random Forest).

E D U C A T I O N

2025 – 2026 | National College of Ireland **MSc Artificial Intelligence**

Thesis: AdaptivePromptGuard (solo) – a two-stage LLM jailbreak defence that cut attack success from up to 77% to under 1% on three 7B models.

Relevant modules: Data Analytics, Machine Learning, Deep Learning, Natural Language Processing, Cloud and Edge AI, Advanced Programming for AI, Research in Computing

2018 - 2022 | Savitribai Phule Pune University **B-Tech Computer Science**

CGPA 8.34/10 | 1st Class with Distinction

I N T E R E S T S A N D A C H I E V E M E N T S

Technical Achievements

- Earned 5-star ratings in Python and SQL on HackerRank.

Fitness & Discipline

- Train regularly at the gym with a focus on strength and endurance – a routine that keeps me disciplined and focused at work.

Travelling

- Moved to Ireland for my MSc and enjoy exploring new cultures and places across Europe, which has broadened how I approach problems and work in diverse teams.